

New Project Ideas - Detailed Build Sheet

SIF Precursor Detection - Reports (remodeled) + CCTV AI Vision + Phone Alerts + Auto-Reports + Site Risk Index + Landing

GOARC Safety 1.0 Prototype - For Every Member, Step by Step (with new auto-report, Risk Index, and Landing)

0 - Remodel Summary

Reports stay but AI drafts the full report from 3 fields (what/where/supervised by) - user only taps Submit or Edit. CCTV page added where a clip is given, AI gives verdict and auto-creates a detailed report. Every change is logged immutably. New: **Auto-generated 9-field incident report** on every CCTV detection + **Site Risk Index** heatmap + **OIL Landing Page** as first page. UI Change - New Landing Page: Old UI opened directly to Dashboard. New UI is **CCTV Surveillance + Report SIF**, so first page (/) becomes a **public informational landing** about the project and how it works for Oil India Company. Dashboard moves to /dashboard. Landing is the story; app is behind Enter Platform. Sections: Hero dark #1A1E22 + blue wash + amber pills (Industrial Safety Dashboard, 600 reports across 15 sites), Why OIL needs this, How it works - 3 cards, For Oil India, Key features grid (auto-report 9 fields, Site Risk heatmap, immutable log, phone push), CTA band #FBBA00.

1 - Auto-Generated Incident Report (New, Detailed)

Trigger: CCTV detects Warning/Serious/Critical -> AI Vision + analyser produce a report automatically with 10 fields. Officer reviews then stores.

What happened: Description from detections: PPE violation + zone intrusion + fall + dwell - e.g., Worker without helmet entered Zone B, then fell

Where it happened: Zone/location/camera: Zone B, Drill Deck, Cam 2, worker foot-point zone lookup

When it happened: Date and timestamp: 30 Aug 2026 14:32:12, server UTC + 30s before context

Why it happened: Possible cause: missing PPE + unauthorized entry + no isolation -> barrier_failure + key_indicators

How it happened: Sequence: enter 14:32:00 -> PPE 14:32:04 -> approach 14:32:08 -> fall 2s 14:32:10-12 -> Critical, stored as JSON

Severity/Risk level: Warning/Serious/Critical via Risk Engine thresholds + confidence ≥ 0.65 + dwell

People/objects involved: Worker 23, helmet No, vest Yes, machinery bbox + class via ByteTrack

Corrective measures: Immediate: isolate area, evacuate, provide PPE, cordon - per LSR Hydrocarbon Release -> isolate leak

Preventive measures: Prevent recurrence: retrain, add barrier, update PTW

CCTV evidence: Snapshot + 30s before/60s after clip + bbox overlay, stored ./data/evidence/

Flow: Detection -> 9-field JSON -> officer Review (Approve/Edit -> re-run AI) -> Store to reports (source=CCTV) + report_logs CCTV_CREATE immutable + push + speaker if Critical.

2 - Site Risk Index (New)

Goal: Visualize riskiest zones so HSE prioritizes. Computed from historical incidents: Frequency (count per zone 90d), Severity (Critical=10, Serious=5, Warning=1), Recency (7d x1.0, 30d x0.6, 90d x0.3). Formula: $RiskIndex = \frac{\sum(severity_weight * recency_decay) * \log(1+frequency)}{10}$ normalized 0-100. Visual: heatmap green->amber->red + ranked Top 3 + trend line. Stored in site_risk_scores (zone, index, frequency, severity_sum, recency, updated_at).

3 - UI Remodel - Landing Page + Report + CCTV + Log

Landing: New **pages/Landing.jsx** (hero + 5 sections + wave dividers), Route "/" = Landing, "/dashboard" = Dashboard (old /), Navbar OIL badge stays, 1300px max-width, Poppins, responsive 1300->900 hamburger. No auth on landing. Reports: 3 fields -> AI draft -> Submit/Edit (risk locked). CCTV: /cctv upload + bbox + zone editor + verdict. Log: report_logs append-only, History timeline read-only.

4 - AI Vision - Detailed (PPE, Zone, Fall etc.)

- PPE Detection: YOLOv8n 8 classes (helmet/vest/gloves/goggles/shoes/harness/mask) 640px conf 0.45, per-zone profile, green/red boxes.
- Worker ID: Badge OCR first (privacy), ByteTrack track_id fallback, map badge->permissions.
- Zone: Polygon ROI 0-1 normalized, Shapely point_in_polygon foot-point, per-camera, red overlay flash.
- Proximity: Worker bbox to machinery ROI distance <50px + dwell >1s -> Serious.
- Fall: Pose flip + dwell >2s + 5-frame window, conf >=0.65 -> Critical.
- Severity: Warning/Serious/Critical table + temporal 2s + confidence.
- Escalation: 60s no ACK -> HSE manager, track ack_at/response_time.
- Alerts: FCM high priority + sound, dashboard red banner + Audio mock.
- Evidence: 30s before/60s after clip + screenshots + JSON.
- Predictive: zone+PPE+proximity -> High-risk warning before fall.

5 - Per-Group Detailed To-Do (with new auto-report, Risk Index, Landing)

App (6 tasks)

Expo + FCM/APNs push_token, 5 screens (Login, Live + auto-report preview, Detail with 9 fields + clip, Settings), JWT, 15s poll fallback, handle Site Risk + Landing deep link.

Frontend (9 tasks)

Simplify Reports 3 fields + AI preview, new /cctv (upload + bbox + zone editor + verdict), new Landing.jsx (hero + 5 sections + wave), auto-report preview modal (9 fields), Site Risk heatmap, History timeline, GOARC, guards.

Backend (9 tasks)

Remove manual risk, add POST /api/cctv/events + /incidents/notify + /acknowledge, auto-generate 9-field report on CCTV (reuse analyser), push fan-out + 60s escalation, compute Site Risk Index nightly -> GET /api/sites/risk.

Database (6 tasks)

report_logs append-only + source enum + push_token + subscriptions, site_risk_scores (zone, index, frequency, severity_sum, recency), indexes, seed --force, Alembic.

AI (10 tasks)

YOLOv8+ByteTrack+pose for auto-report 9 fields + Site Risk calc, leakage boost already done, fire classifier, zone point-in-poly, temporal 2s, severity thresholds, lazy load.

Docker/DevOps (5 tasks)

ai-vision 8002 (libgomp1+ultralytics), compose 8002 + FCM + CCTV_RTSP, start.sh warm-up, .dockerignore.

Git (3 tasks)

feat/report-remodel, feat/cctv-auto-report, feat/site-risk-index, feat/landing-page, feat/mobile-push -> tag v0.5

QA/Demo (4 tasks)

2 canned clips -> auto 9-field report push -> ACK -> History + heatmap + landing demo shows Zone B red.